

Wireless Transmission

Energy Transmission
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Content

- ◆ History & Principles of Wireless Power
- ◆ Transmission Methods: Inductive, Capacitive, Far-Field
- ◆ Standards & Key Applications (Consumer, EVs, Medical)
- ◆ Safety, Standards & Environmental Impact
- ◆ Challenges & Future Outlook

Basic Principles of Wireless Power

- ♦ Wireless electricity transmits energy without physical conductors using oscillating magnetic or electric fields.
- ♦ Main techniques: inductive coupling, resonant inductive coupling, capacitive coupling, and far-field electromagnetic radiation.
- ♦ **Inductive coupling** uses coils to generate near-field magnetic flux for short-range energy transfer.
- ♦ **Resonant inductive** coupling tunes coils to same resonant frequency, enhancing efficiency and range.
- ♦ **Capacitive coupling** uses electric fields between conductive plates, effective at low power and short range.
- ♦ **Far-field methods** (microwaves, RF) enable longer distance power transfer but need line-of-sight and safety considerations.

Conductive Power Transmission

A conductively powered tram (left) and its pantograph (right), which is one of the oldest types of mobile power transmission.

→ Not “wireless”



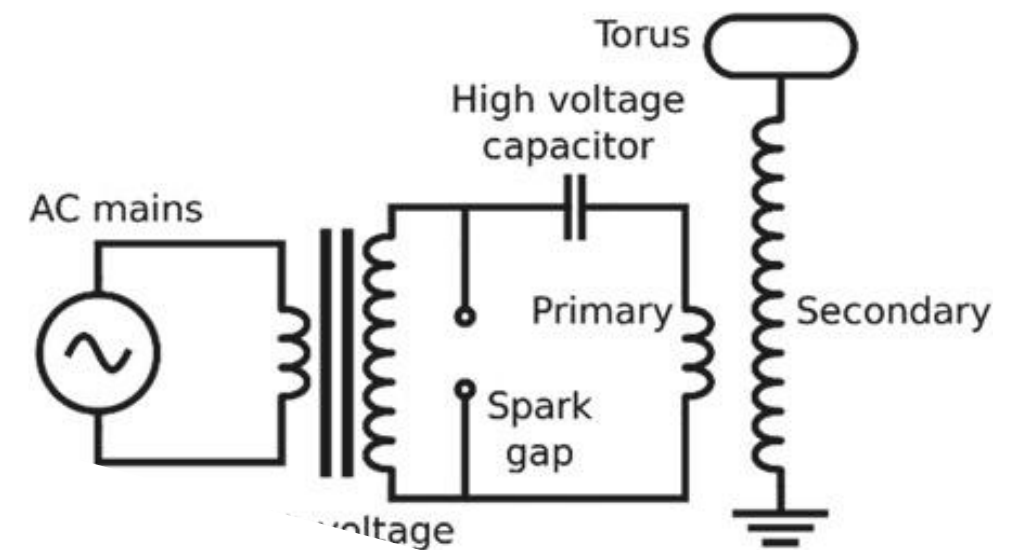
History of Wireless Power

- ◆ Origins trace back to 19th century with Michael Faraday and James Clerk Maxwell, who developed electromagnetic theory.
- ◆ Nikola Tesla's 1890s experiments with Tesla coil and Wardenclyffe Tower demonstrated wireless energy transmission concepts.
- ◆ Tesla's vision aimed at global wireless power distribution but was never fully realized due to funding and technical challenges.
- ◆ Wireless power interest diminished mid-20th century with wired power dominance, but resurged in the 21st century with new technological applications



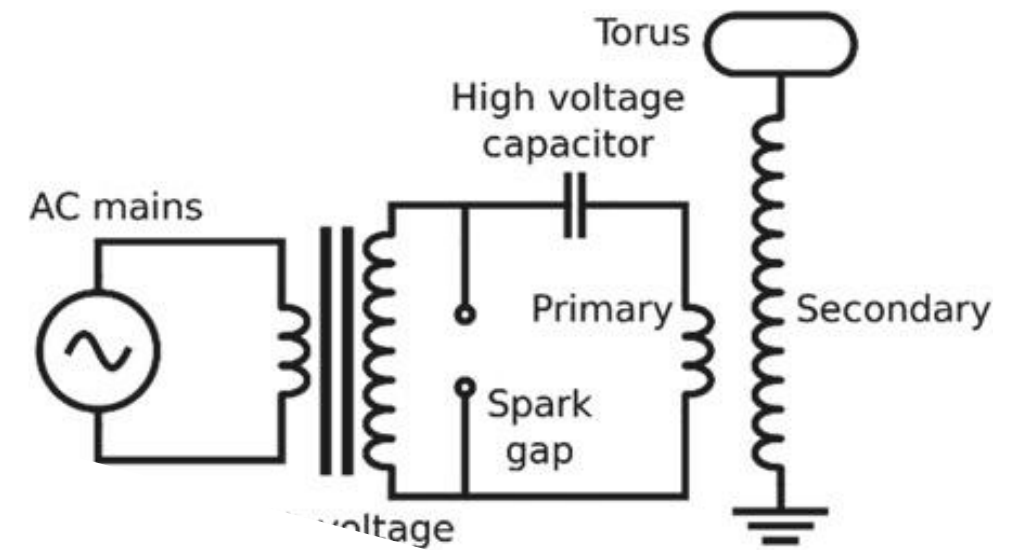
Tesla coil (history)

- In 1891, Tesla patented the **Tesla coil**, a device using **resonant inductive coupling**
- A Demonstrated the basic principle of magnetic resonance wireless power transfer (WPT)
- Tesla's work laid the foundation for modern wireless charging, though it was not commercialized at the time



Tesla coil (principle)

- The system consists of a primary coil, a secondary coil, and a capacitor.
- The capacitor charges and then discharges through the primary coil, creating a magnetic field.
- This magnetic field induces a current in the secondary coil through electromagnetic induction.
- The result is a buildup of high voltage in the secondary coil, producing visible electrical arcs.



Tesla coil (principle)

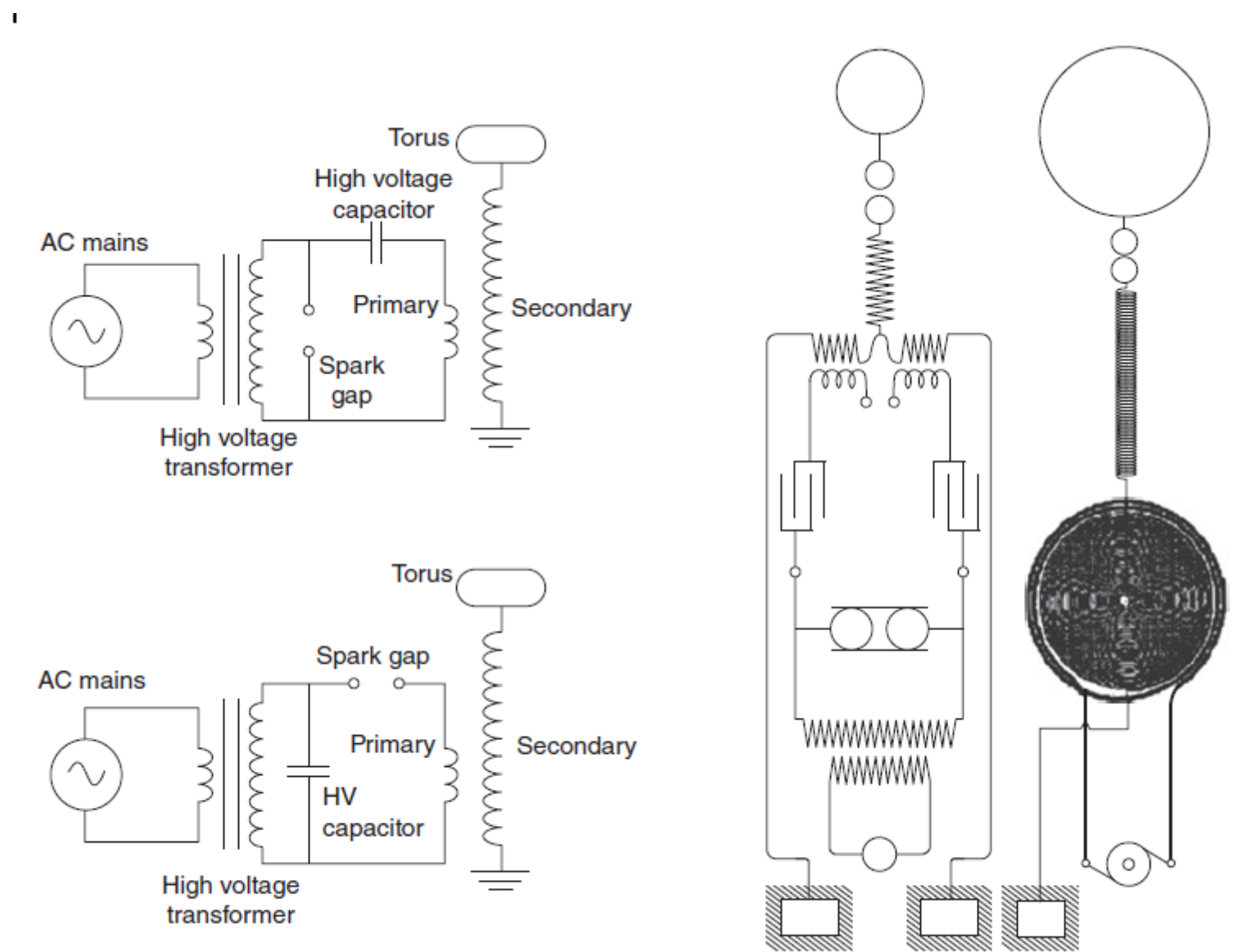


Figure 1.4 Examples of the Tesla coil, which is a type of resonant transformer (25 kHz–2 MHz), invented in 1891.

Basic Principles of Wireless Power

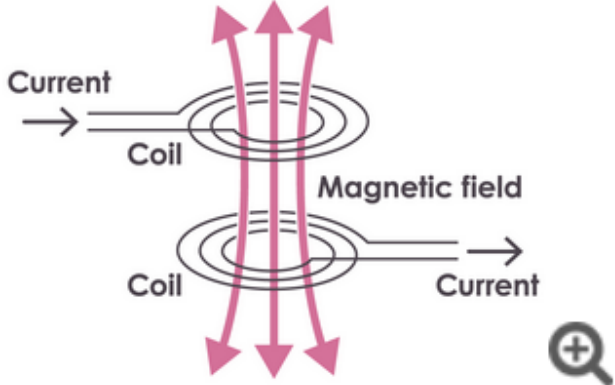
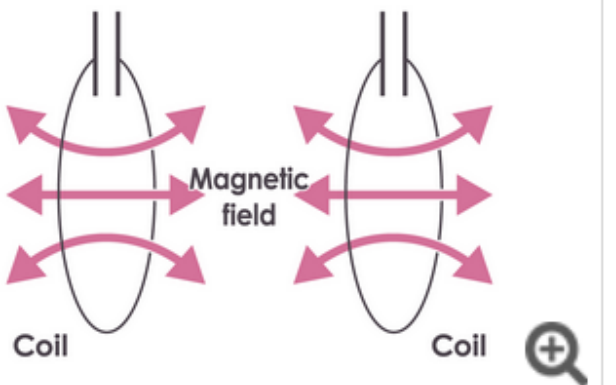
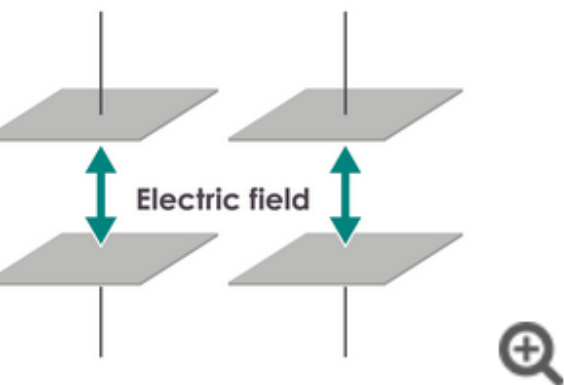
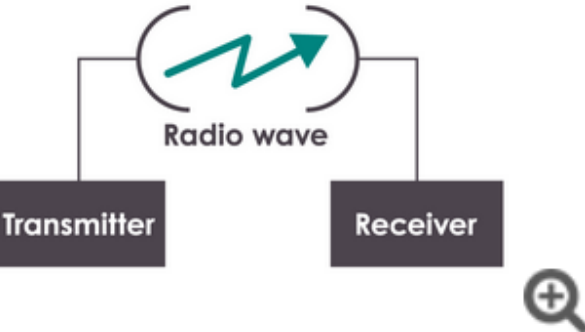
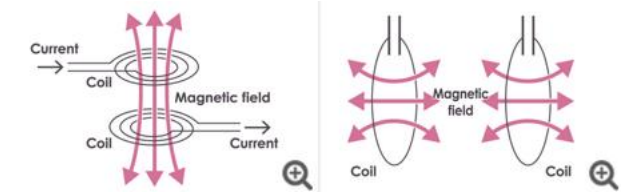
Classification	Coupling method			Radiation method
	Electromagnetic induction		Electric field coupling	Microwave
	Magnetic coupling (Magnetic field coupling)	Magnetic field resonance		
Image				
Advantage	- High efficiency in power transmission - Stable transmission	- Allows relatively long-distance transmission	- Enables high power transmission - High efficiency in power transmission	- Allows long-distance transmission - Supports multiple receivers
Drawback	- Limited distance	- Sensitive to circuit constants	- Requires high voltage - Very limited distance	- Low efficiency in power transmission
Use case	- Electric toothbrushes - Qi standard - ABLIC's wireless power transfer	- AGV (Automated guided vehicle) - EV charging		- IoT sensors

Image source:
<https://www.ablic.com/en/semicon/products/rtc/wireless-power-ic/intro/what-wireless-power/> -

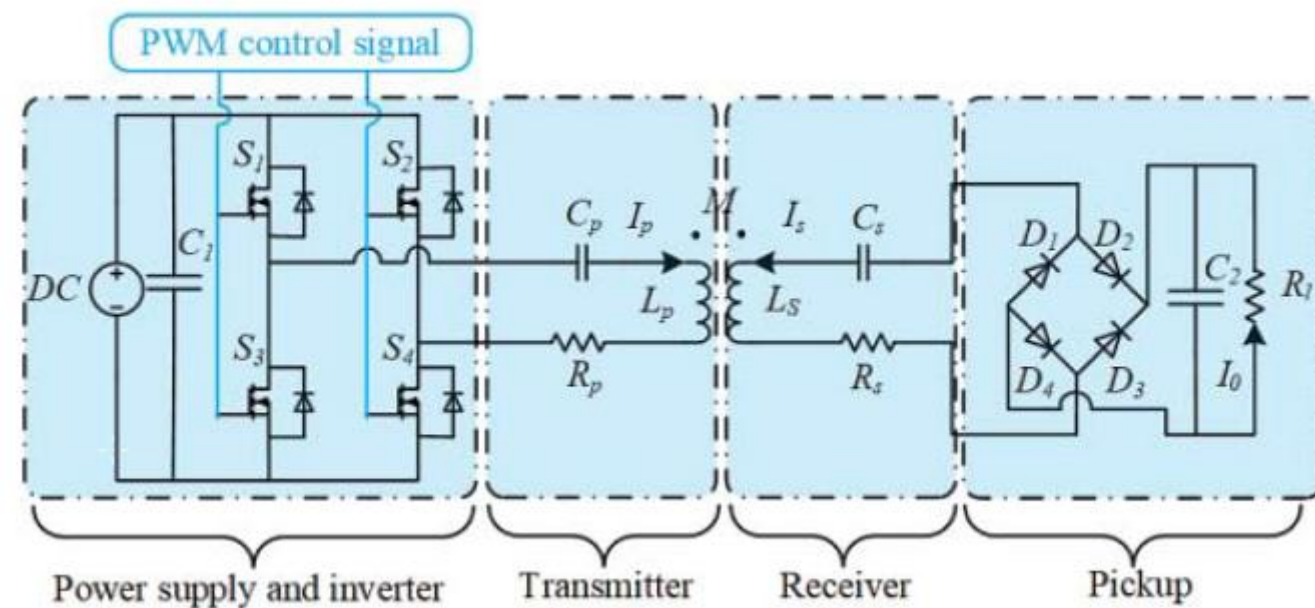
Inductive Coupling and Resonant Inductive Coupling

- ◆ Inductive coupling is widely used for charging smartphones and small devices, relying on magnetic fields between closely spaced coils.
- ◆ Highly efficient at short distances but sensitive to coil alignment and distance variations.
- ◆ Resonant inductive coupling improves range and efficiency by matching coil resonance frequencies, allowing power transfer over several centimeters to meters.
- ◆ Used increasingly in electric vehicle wireless charging and medical implants for greater spatial freedom and tolerance.

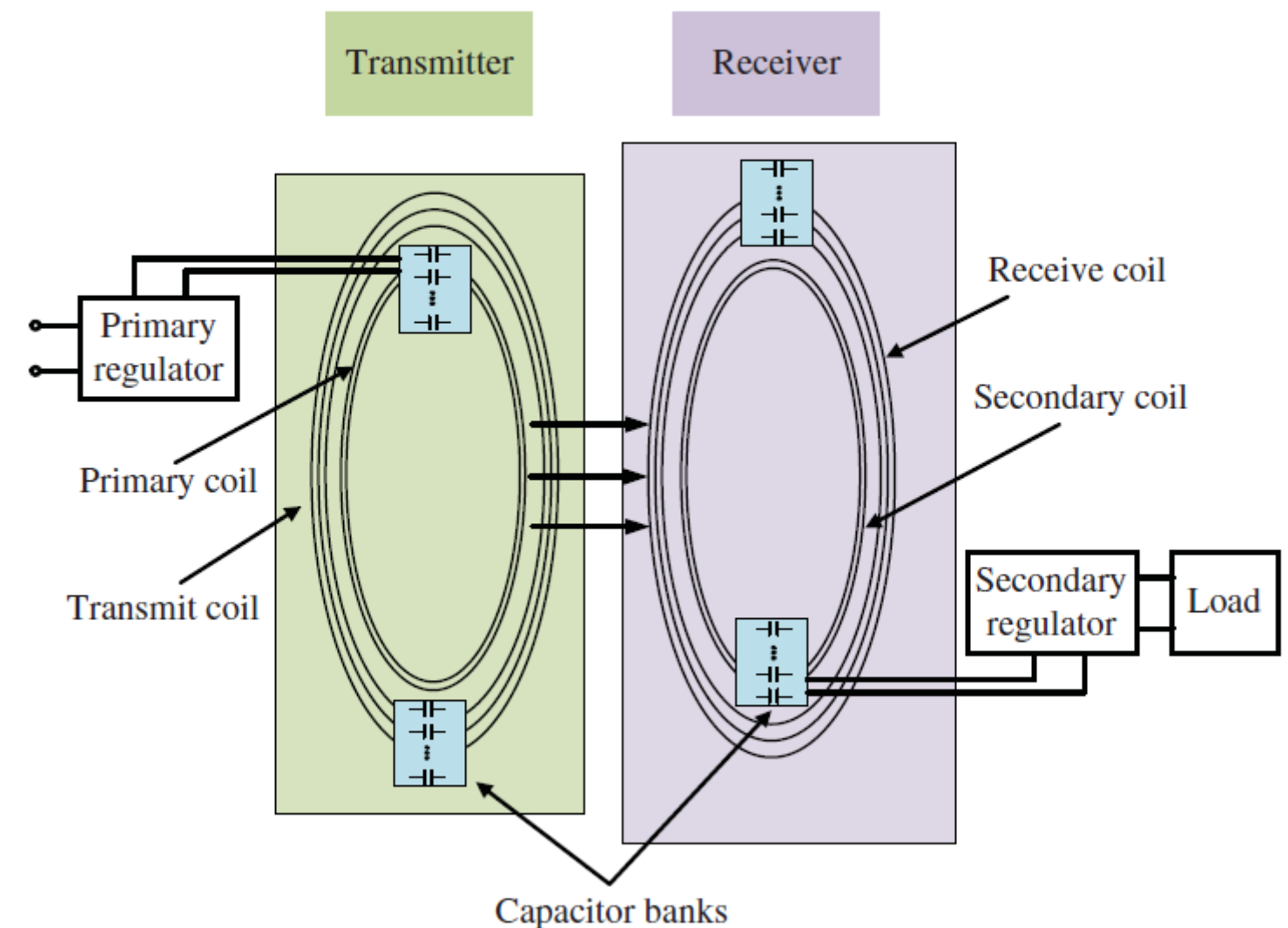


Inductive coupling

Simple:



Resonant coupling:



At resonance, the system naturally stores and returns energy between the capacitor and coil (in an LC circuit), which reduces resistive and radiative losses.

Capacitive Coupling

- ♦ Capacitive coupling transfers energy via electric fields between conductive plates separated by a dielectric.
- ♦ Suitable for low-power applications like biomedical implants and sensors, but sensitive to environmental factors.

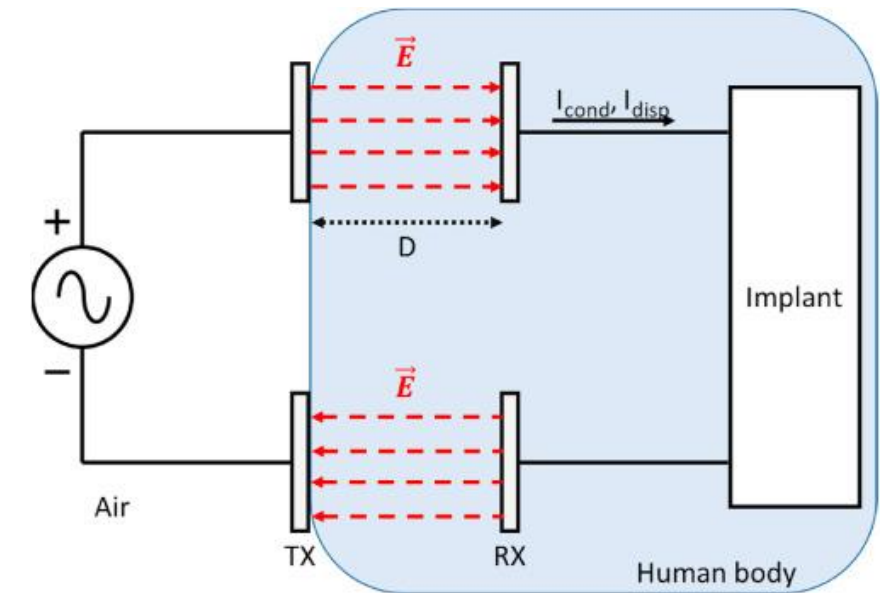


Image source: Sciencedirect

Capacitive Coupling

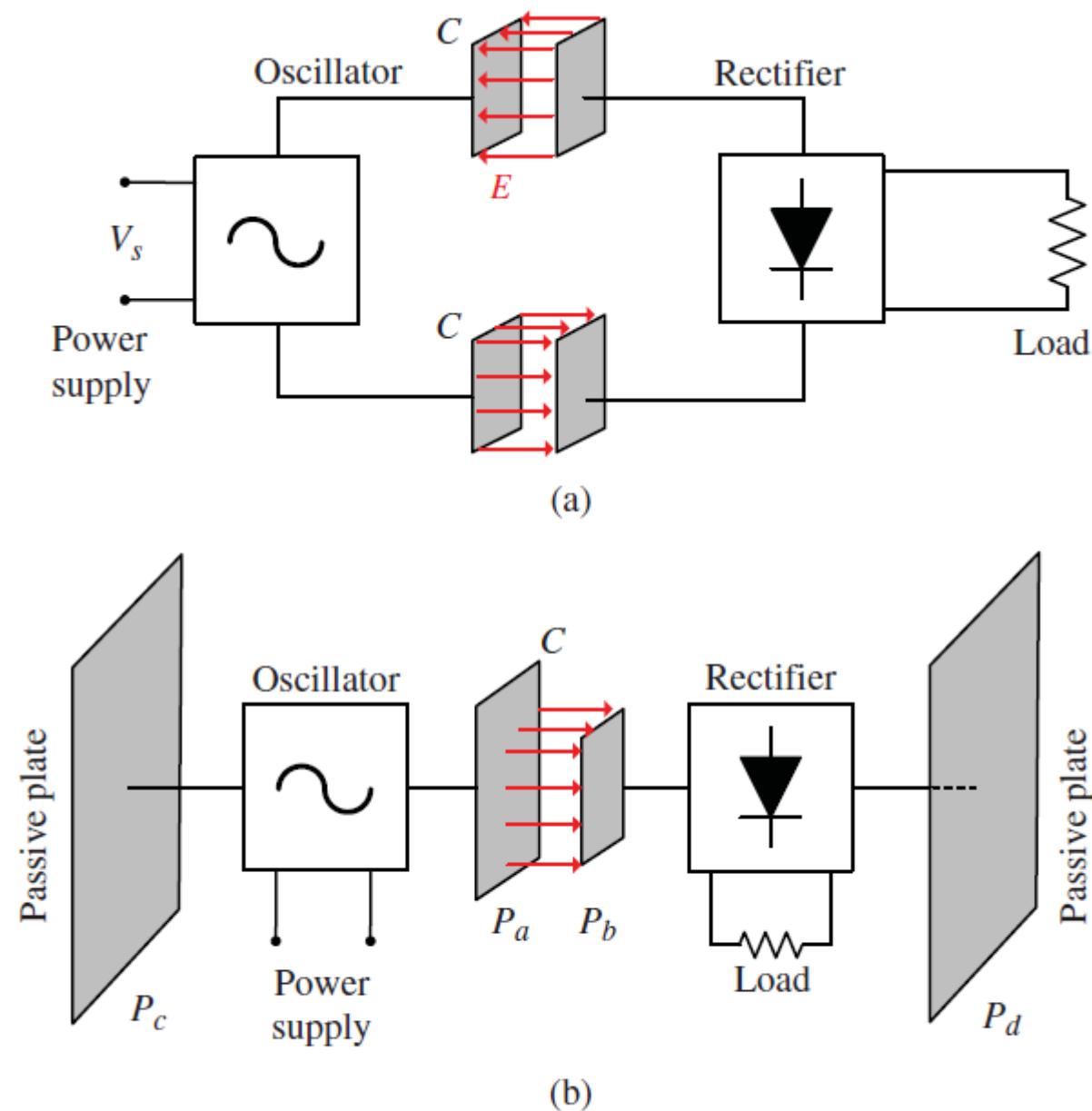


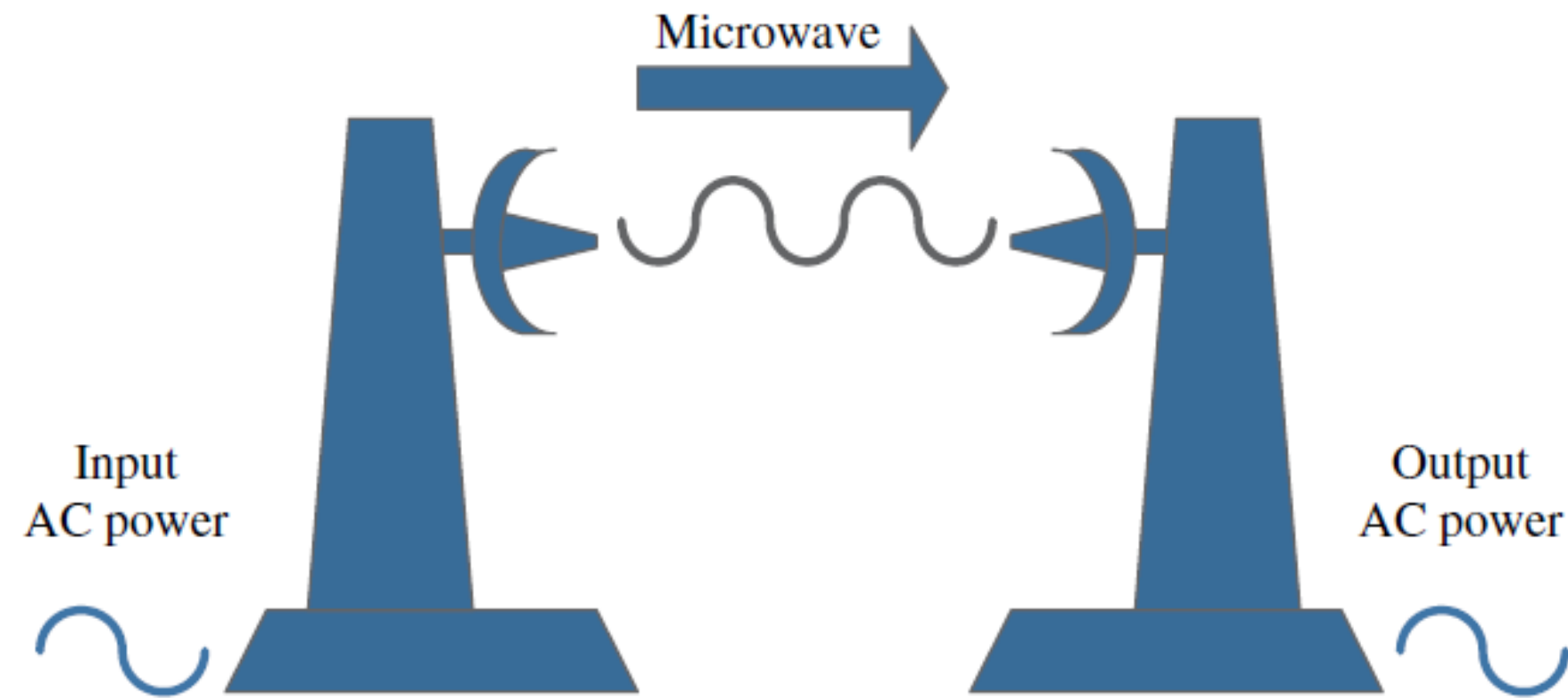
Figure 1.7 Capacitive wireless power transfer: (a) Bipolar; (b) Unipolar.

- **Bipolar CPT** is preferred for:
 - Higher power applications (e.g. industrial sensors)
 - Scenarios requiring better EMI control and coupling
 - Disadvantage (more complex circuitry)
- **Unipolar CPT** is suitable for:
 - Low-power devices (e.g., sensors, wearables)
 - Cost-sensitive or compact designs

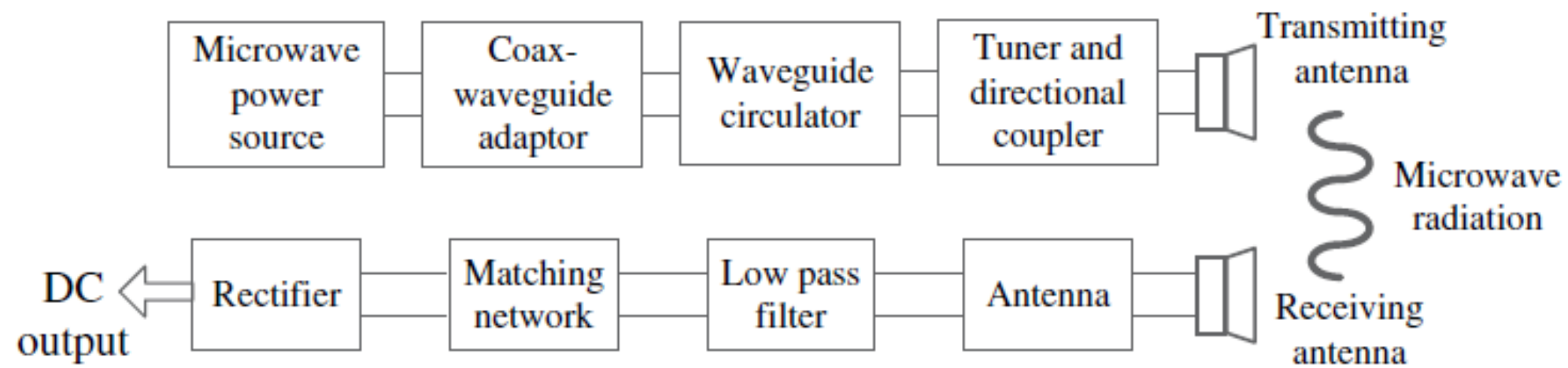
Far-Field Methods: Microwave & Radio Frequency

- ◆ Microwave Transmission: Transfers power over long distances. Actively explored for Space Solar Power (SSP) systems, enabling continuous renewable energy beaming from orbit to Earth.
- ◆ Radio Frequency (RF) Transmission: Powers small devices (e.g., IoT sensors, RFID tags). Practical implementation is constrained by regulatory limits on power/spectrum and efficiency challenges.

Microwave transmission

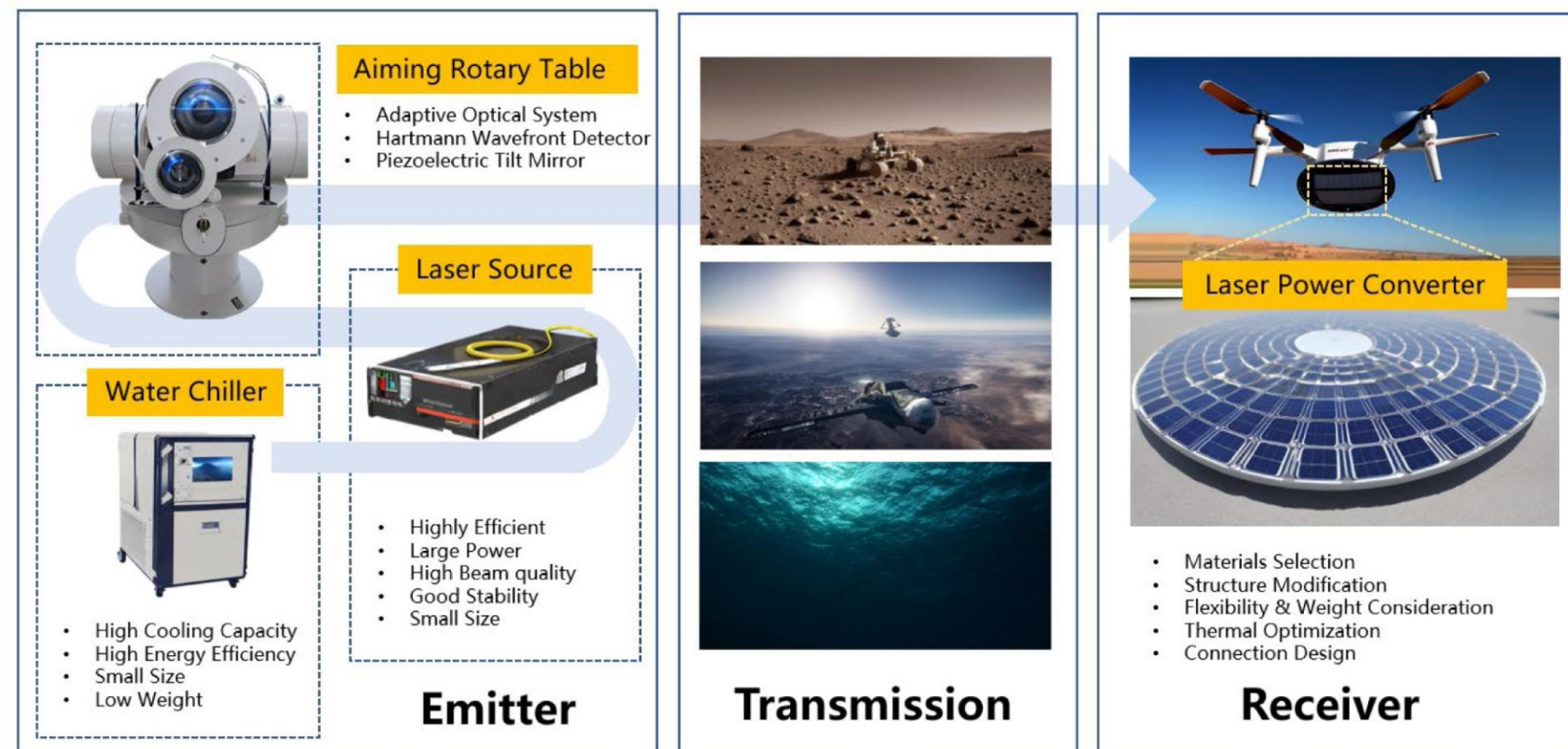


(a)



Far-Field Methods: Laser

- ♦ Laser Transmission: Uses focused light beams for highly efficient, long-distance power transfer. Excelling in applications requiring precision (e.g., powering drones, space-to-space links), it faces challenges from atmospheric absorption/scattering and requires strict safety protocols.



Consumer Electronics Applications

- ♦ Wireless charging based on inductive charging prevalent in smartphones, smartwatches, earbuds, and personal devices
- ♦ Eliminates cables, improves convenience
- ♦ Supported by charging pads, furniture integration, and public charging stations worldwide.
- ♦ Ongoing improvements aim for faster charging, wider compatibility, and longer ranges.

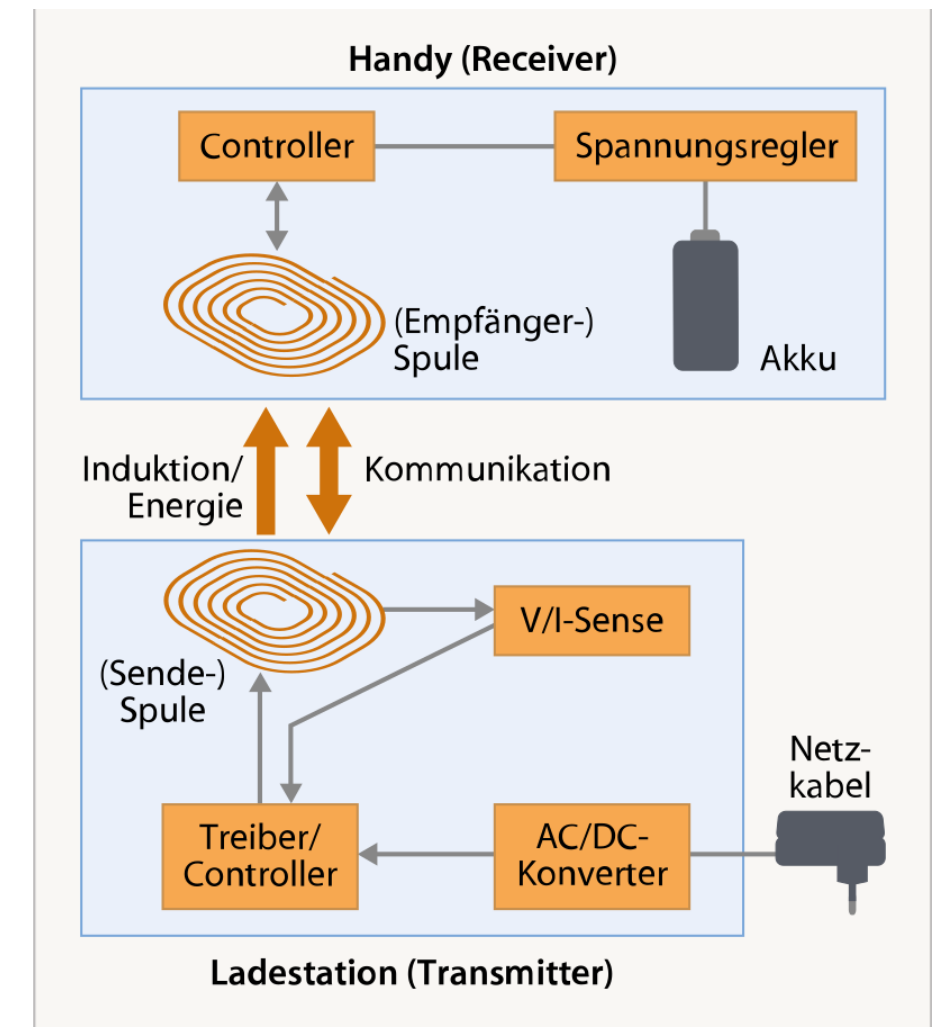
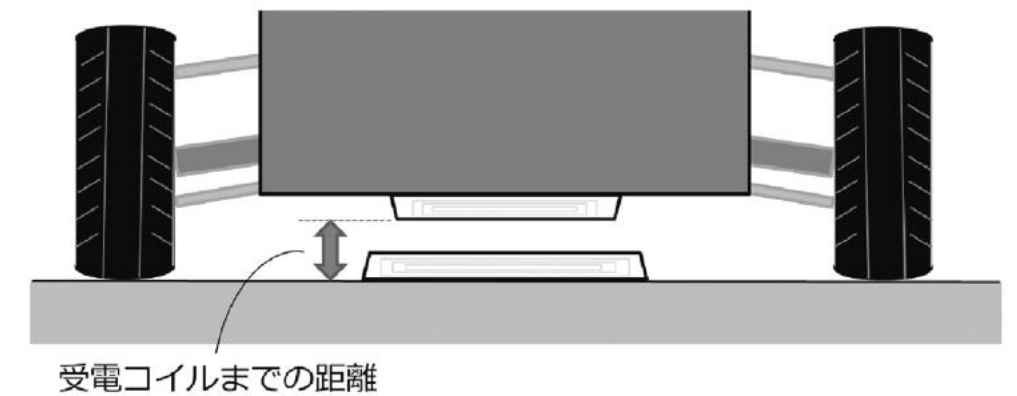


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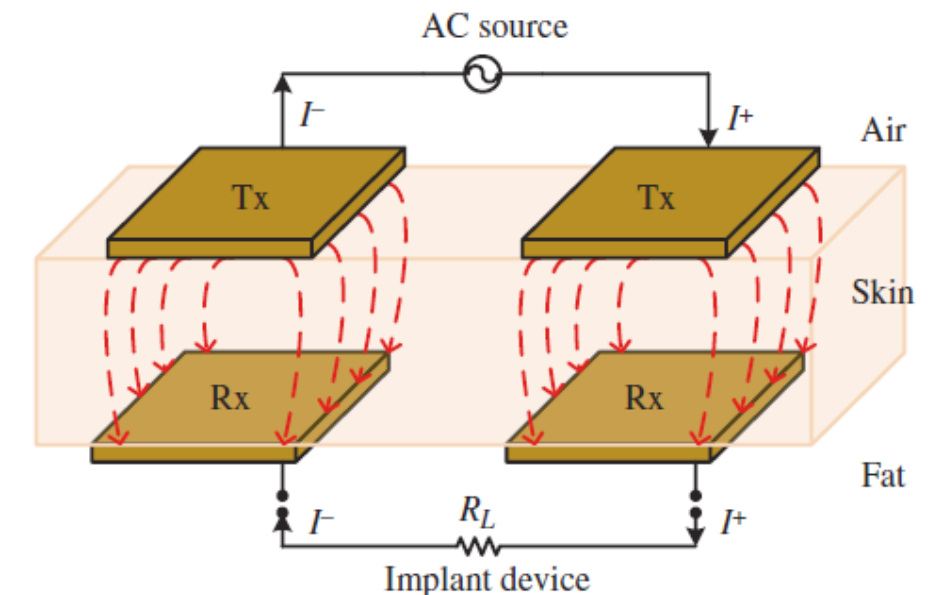
Electric Vehicle (EV) Charging

- ◆ Resonant inductive charging pads embedded in garages, parking lots, and roads provide convenient EV charging.
- ◆ Power levels range from a few kW up to 22 kW or more, enabling faster, cable-free charging.
- ◆ Enhances safety and reduces maintenance by eliminating physical connectors.
- ◆ Pilot projects worldwide validate technology, with future prospects including dynamic charging while driving.



Medical and Implantable Devices

- ◆ Wireless power critical for implantable devices like pacemakers and neurostimulators, reducing infection risk by avoiding wires.
- ◆ Capacitive and inductive coupling commonly used depending on device size and power needs.
- ◆ Stringent safety and reliability standards ensure patient safety and device longevity.
- ◆ Research focuses on enhancing efficiency and miniaturization to improve patient outcomes.



Industrial and Automation Uses

- ♦ Wireless power enables continuous operation of robots, drones, and sensors without cable management downtime.
- ♦ Reduces maintenance costs and cable wear, enhancing factory safety and flexibility.
- ♦ Adopted in harsh environments where cables degrade rapidly or hinder mobility.
- ♦ Supports Industry 4.0 goals of flexible, automated production systems.



Wireless Charging Roads and Dynamic Charging

- ◆ Charging infrastructure embedded in roadways can power EVs while driving, extending range and reducing battery size.
- ◆ Pilot projects in Sweden, South Korea, and Israel demonstrate technology feasibility.
- ◆ Dynamic charging requires precise coil alignment and durable infrastructure materials.
- ◆ Potential to revolutionize transportation by integrating power delivery into road networks.

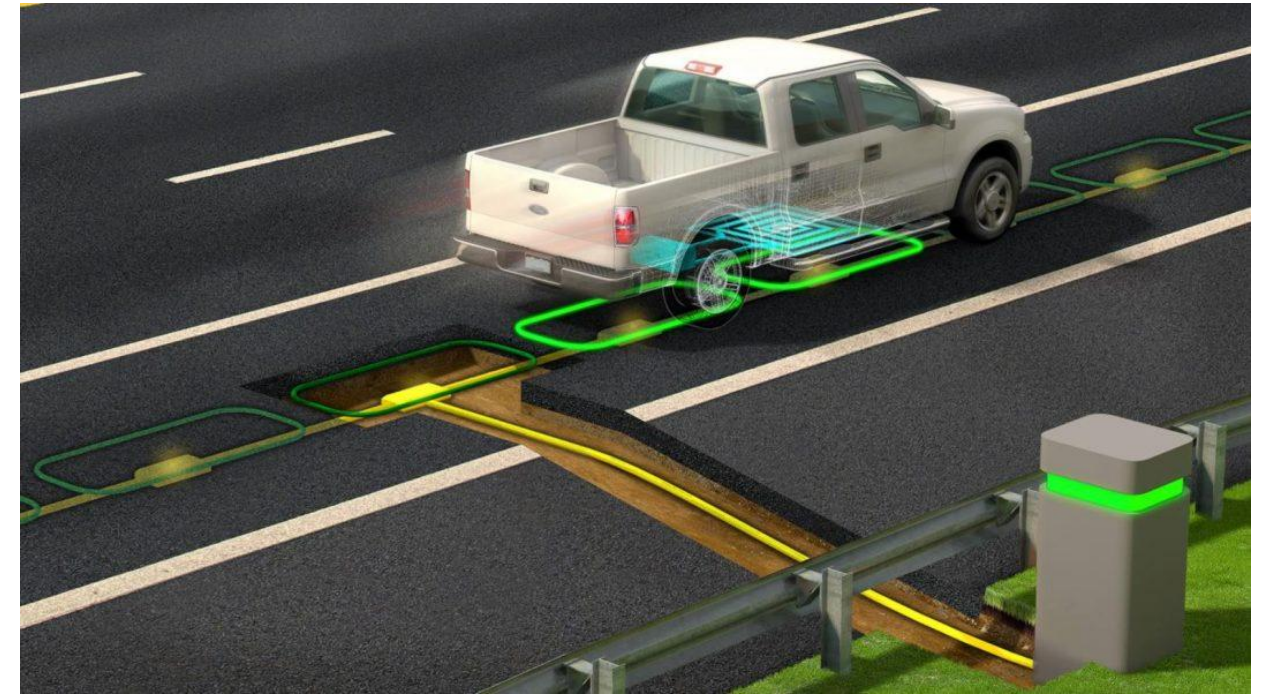
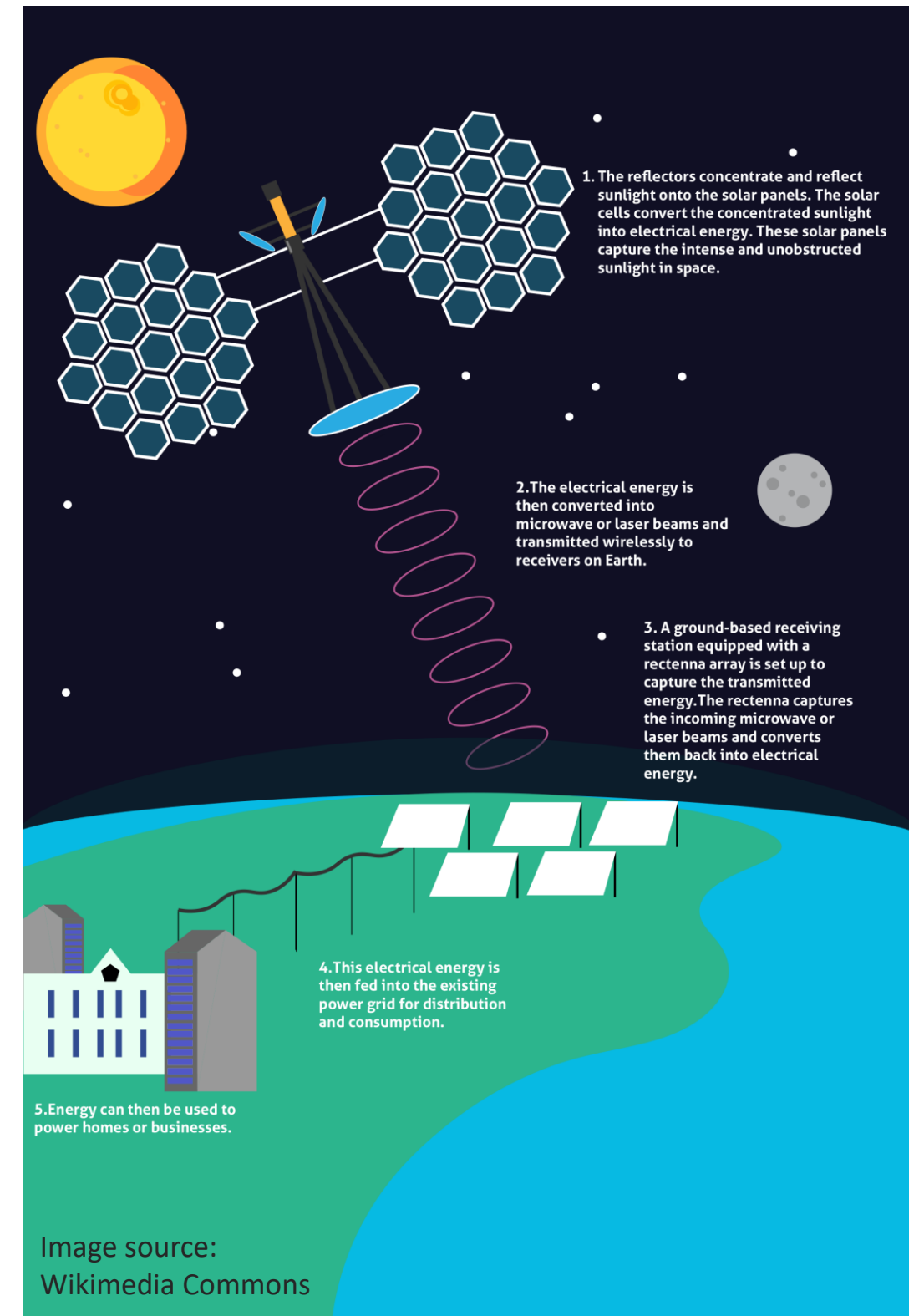


Image source:
eletimes.com

Space-Based Wireless Power

- ◆ Space solar power systems collect sunlight in orbit and beam energy to Earth via microwaves.
- ◆ Offers clean, continuous power regardless of weather or time of day.
- ◆ Challenges include launch costs, beam focusing, and safety protocols.
- ◆ NASA and international agencies actively research feasibility and technology development.



Efficiency and Losses in Wireless Power

- ◆ Efficiency varies with distance, coil alignment, and environmental factors.
- ◆ Near-field methods typically achieve 70-90% efficiency at close range.
- ◆ Far-field methods have lower efficiency due to spreading losses and atmospheric absorption.
- ◆ Balancing efficiency with safety and cost is critical for commercial success.

Safety Considerations

- ◆ Wireless power transmission must comply with electromagnetic exposure limits to ensure human safety.
- ◆ Near-field techniques generally have minimal health risks due to low power density and short ranges.
- ◆ Far-field methods require strict beam control and safety interlocks to prevent accidental exposure.
- ◆ Standards organizations provide guidelines for safe wireless power system design and operation.

Standards of wireless power transfer

Table 1.2 Different standard of WPT.

Standard	Title (Content)	Application
Recommendation ITU-R SM.2110	Guidance for the use of frequency ranges for operation of non-beam wireless power transmission for electric vehicle	EV
Report ITU-R SM.2303-2	Wireless power transmission using technologies other than radio frequency beam	All
Recommendation ITU-R SM.2110	Frequency ranges for operation of non-beam Wireless Power Transmission systems	All
IEC61980-1	Electric vehicle wireless power transfer (WPT) systems - Part 1: General requirements	EV
IEC61980-2	Electric vehicle wireless power transfer (WPT) systems - Part 2: Specific requirements for communication between electric road vehicle (EV) and infrastructure	EV
IEC61980-3	Electric vehicle wireless power transfer (WPT) systems - Part 3: Specific requirements for the magnetic field wireless power transfer systems	EV
ISO 19363:2017	Electrically propelled road vehicles? Magnetic field wireless power transfer? Safety and interoperability requirements	EV
ISO 19363:2020	Electrically propelled road vehicles? Magnetic field wireless power transfer? Safety and interoperability requirements	EV
SAE J2954	Wireless Power Transfer for Light-Duty Plug-in/Electric Vehicles and Alignment Methodology	EV
SAE J2847/6	Communication for Wireless Power Transfer Between Light-Duty Plug-in Electric Vehicles and Wireless EV Charging Stations	EV
Qi standard	The world's de facto wireless charging standard for providing 5-15 watts of power to small personal electronics	Phone

Environmental Impact

- ♦ Wireless power can reduce electronic waste by extending device lifetimes and eliminating connectors.
- ♦ Potential for reduced resource use by enabling more efficient energy transfer in smart grids and EVs.
- ♦ Challenges include energy losses during transmission and material use for coils and infrastructure.
- ♦ Lifecycle analyses help guide sustainable design and deployment.

Cost and Economic Impact

- ◆ Initial costs for wireless power systems are higher due to specialized hardware and infrastructure.
- ◆ Long-term savings come from reduced maintenance, improved device longevity, and convenience benefits.
- ◆ Market for wireless charging expected to grow rapidly with EV and IoT adoption.
- ◆ Economic barriers include standardization, consumer awareness, and infrastructure costs.

Summary and Outlook

- ◆ Wireless electricity is transitioning from niche to mainstream with broad applications across industries.
- ◆ Advances in materials, electronics, and standards are driving improvements in performance and adoption.
- ◆ Future holds promise for wireless power integrated seamlessly into daily life and infrastructure.

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